IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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07.08.2025

Outline

Executive Summary

Methodologies Overview

- **Data Collection**
  - Retrieved detailed launch, core and payload data via the SpaceX REST API
- **Web Scraping**
  - Scraped historical Falcon 9 & Heavy launch tables from Wikipedia for outcome labels
- **Exploratory Data Analysis & Visualization**
  - Explored booster version, payload mass, flight count and orbit impacts on landing success
  - Visualized correlations and distributions to inform feature engineering
- **Geo-Spatial Analysis**
  - Mapped launch pads with Folium and quantified site-specific landing success rates
- **Visual Analytics**
  - Performed interactive visual analytics using Plotly Dashboard.
- **Model Building & Training**
  - Evaluated Logistic Regression, SVM, Decision Tree and KNN on an 80/20 train/test split
- **Hyperparameter Tuning & Cross-Validation**
  - Performed 10-fold GridSearchCV to optimize LR, SVM, Decision Tree and KNN parameters

Key Results Summary

- **Best Model:** Logistic Regression achieved **83.3% test accuracy** with perfect recall on successful landings, ensuring no true successes were missed.
- **Key Predictors:** Core reuse count and prior flight number had the highest coefficients, with payload mass, orbit type, and launch site also significantly influencing outcomes.
- **Operational Insights:** KSC LC-39A leads in landing reliability (~90%), and missions in the **7 500–10 000 kg** payload band show the strongest recovery performance.

Introduction

- **Project Background & Context**
    - Part of IBM Data Science Professional Certificate capstone, focused on SpaceX first-stage landing prediction.
    - Leverages public SpaceX REST API data combined with web-scraped historical launch records.
    - Goal: Build a data-driven model to predict Falcon 9 landing success and support mission planning.

- **Problems & Questions to Answer**
    - **Can we predict a booster's landing outcome** using pre-launch features (payload mass, booster version, prior flights)?
    - **Which factors** (core reuse count, flight history, orbit type, launch site) most strongly influence landing success?
    - **What is the optimal modeling approach** (algorithm selection and hyperparameters) for this classification task?
    - **How consistent are landing success rates** across different launch pads, and can spatial features improve predictions?

Section 1

Methodology

Methodology

Executive Summary

- Data Collection methodology
  - Request to the SpaceX API
  - Web Scraping Falcon 9 launch record from Wikipedia
- Perform data wrangling
  - Converting mission outcomes into Training Labels
    - 1 means the booster successfully landed
    - 0 means it was unsuccessful
- Perform EDA using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Standardize the data
  - Split into train and test
  - Find the best Hyperparameters for SVM, Classification Trees and Logistic Regression
    - Find the method that generalizes the best

Data Collection

Objective: Gather raw launch data to build the foundation for first-stage landing prediction.

Data Collection Process:

- Data Collection using SpaceX API- Fetching data from Static URL using Get Request.
- Data Collection using Web Scraping- Scraping data from HTML tables on Wikipedia.

Key Endpoints & Functions:

- **getBoosterVersion(data):** Retrieves rocket configuration details from /rockets/{id}
- **GetLaunchSite(data):** Retrieves the name of the launch site being used, the longitude, and the latitude
- **getPayloadData(data):** Fetches payload mass and orbit info from /payloads/{id}
- **getCoreData(data):** Pulls core reuse, landing success, and flight history from /cores/{id}

Data Assembled:

- Combined into a single DataFrame with columns for booster version, payload mass, core reuse count, and prior landings
- Ready for cleaning and feature engineering steps

Data Collection – SpaceX API

Data Collection Steps:

1. Get request to the SpaceX API.
2. Conversion of JSON Response into pandas dataframe.
3. Get Information about the launches using the IDs given for each launch.
4. Calling getBoosterVersion, getLaunchSite, getPayloadDataand getCoredata methods to obtain the data.
5. Apply Data Filtering to only include Falcon 9 launches.
6. Handle missing values.
7. Generate a CSV file.

(Github Link): https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/1_jupyter-labs-spacex-data-collection-api.ipynb

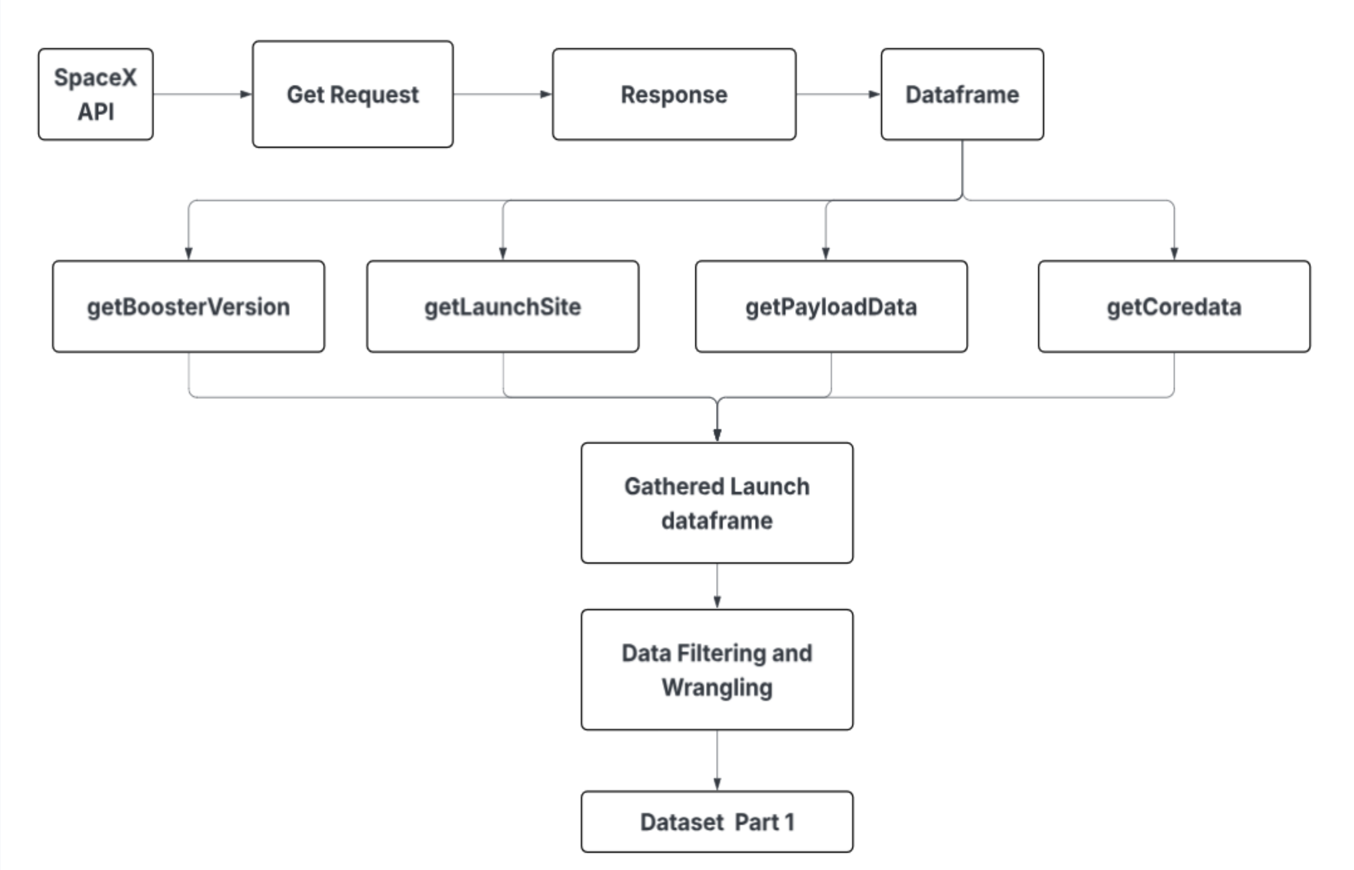


Figure.1 Flowchart of the Data Collection process using SpaceX API

Data Collection - Scraping

Web Scraping Process Steps:

1. Get Request to the Wikipedia Page.
2. Conversion of Response into BeautifulSoup Object.
3. Extraction of all column names from table 3.
4. Parsing launch HTML tables to generate dataframe
5. Exporting as a CSV File.
6. GitHub URL : (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/2_jupyter-labs-webscraping.ipynb)

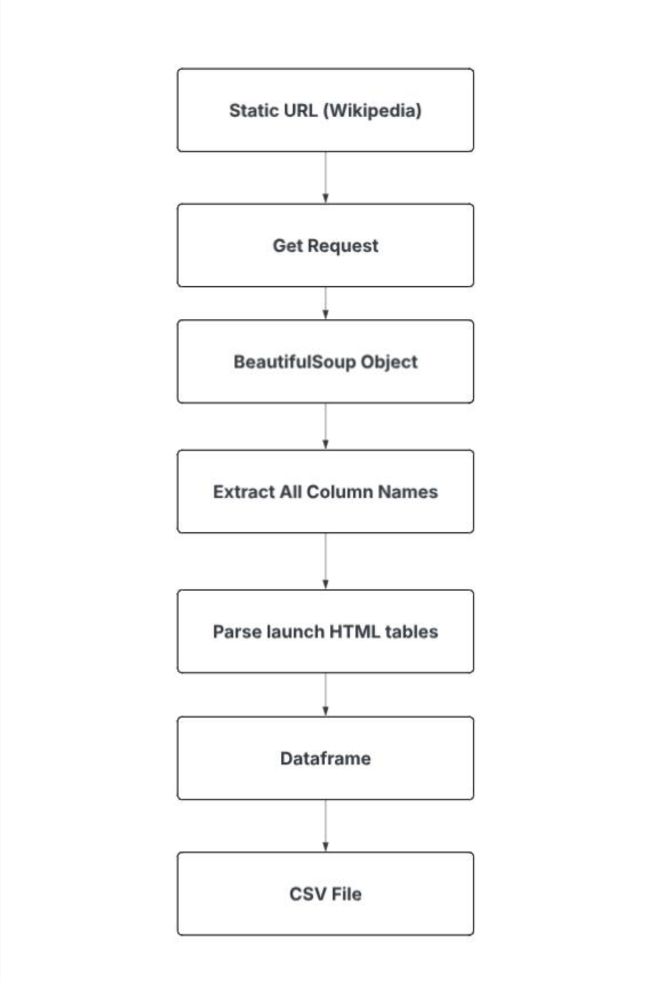


Figure.2 Flowchart of the Data Collection - Scraping

Data Wrangling

Data Wrangling Process:

1. Load the dataset and perform basic checks.
2. Calculate Number of launches per site.
3. Calculate Number and Occurrence of each Orbit.
4. Calculate number and outcomes of Mission outcome of the orbits.
5. Generate targate class variable from outcome column.
6. Export the dataset as csv file.
7. [GitHub URL](https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/3_labs-jupyter-spacex-Data%20wrangling.ipynb) : (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/3_labs-jupyter-spacex-Data%20wrangling.ipynb)

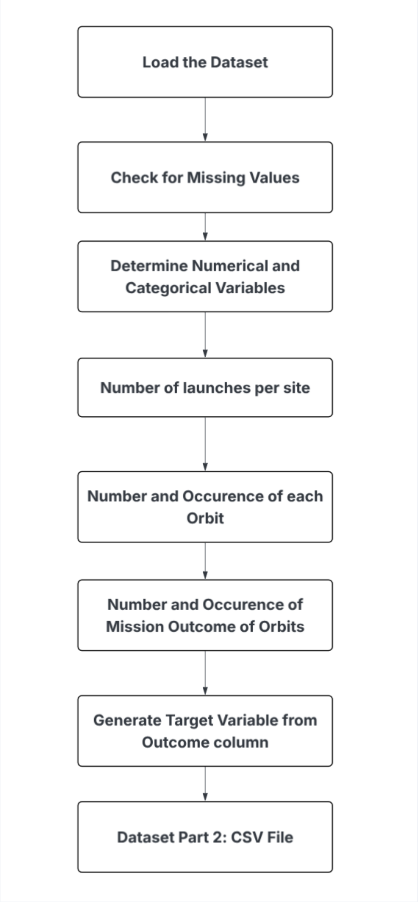


Figure.3 Flowchart of the Data Wrangling Process

EDA with Data Visualization

Key EDA Visualizations (Plots)

1. Flight Number vs. Launch Site (Scatter Plot)

- Visualizes mission cadence by pad and highlights site-specific early failures vs. later successes.

2. Payload Mass vs. Launch Site (Scatter Plot)

- Shows which pads handle heavier or lighter payloads—suggesting interaction of mass & site on recovery.

3. Flight Number vs. Orbit Type (Scatter Plot)

- Reveals how booster experience aligns with different orbit profiles.

4. Payload Mass vs. Orbit Type (Scatter Plot)

- Illustrates orbit-specific mass ranges, since high-energy orbits often carry particular payload weights.

5. Success Rate by Orbit Type (Bar Chart)

- Compares landing recovery percentages across orbits, spotlighting inherently easier vs. harder missions.

6. Yearly Success Rate (Line Chart)

- Tracks average first-stage landing rate over calendar years, evidencing SpaceX's learning curve.

7. Flight & Payload vs. Outcome (Colored Scatter Plot)

- Plots flight number vs. payload mass colored by success/failure, highlighting regions where well-flown cores + moderate loads yield the best recoveries.

These seven visuals together guide feature engineering by pinpointing the strongest individual and joint predictors of landing success.

Github URL: (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/5_edadataviz.ipynb)

EDA with SQL

SQL Queries Summary

- **Prepared Working Table**
  - Created SPACEXTABLE filtering out records with null dates
- **Data Preview & Exploration**
  - Viewed first 10 rows to inspect schema and contents
  - Retrieved distinct launch sites for site-based feature insight
- **Basic Filters**
  - Selected Cape Canaveral launches (Launch_Site LIKE 'CCA%')
  - Isolated missions in 2015 (SUBSTR(Date,1,4) = '2015')
- **Aggregations & Statistics**
  - Calculated total payload mass for NASA CRS missions
  - Computed average payload mass for booster version "F9 v1.1"
  - Found earliest landing success date
  - Counted missions by landing outcome
- **Advanced Analysis**
  - Identified booster versions whose total payload mass exceeds the dataset average
  - Tallied launch counts per site since March 20, 2017, ordered by frequency

- Github URL : (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/5_1_jupyter-labs-eda-sql-coursera_sqllite.ipynb)

Build an Interactive Map with Folium

Folium Map Objects & Their Purpose

- **Circle Markers**
  - Placed at each launch site's latitude/longitude
  - **Why:** Show precise pad locations; size of each circle scaled by total mission count to convey operational volume
- **Color-Coded Popups / Tooltips**
  - Each marker includes a popup (or tooltip) displaying the site name and its landing success rate
  - **Why:** Allow interactive inspection of site-specific performance without cluttering the map
- **Legend (Custom Control)**
  - Added a legend overlay mapping colors (e.g. green → high success, red → lower success)
  - **Why:** Helps viewers immediately interpret success-rate shading on the map
- **Base Map Layers**
  - Included standard tiles (e.g. OpenStreetMap) and optional satellite imagery layer
  - **Why:** Gives context (terrain, infrastructure) and lets users toggle for clearer geographic orientation
- **Layer Control Widget**
  - Enables toggling of marker layer and any additional overlays
  - **Why:** Empowers users to focus on the geospatial data of interest (e.g. hide/show success-rate circles)

Github URL: (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/6_lab_jupyter_launch_site_location.ipynb)

Build a Dashboard with Plotly Dash

Dashboard Components & Interactions

- **Launch Site Dropdown** (**Type:** dcc.Dropdown)
  - **Purpose:** Let users filter all visuals by a specific launch site or view “All Sites” at once.
- **Payload Range Slider (Type:** dcc.RangeSlider)
  - **Purpose:** Enable dynamic filtering of missions by payload mass (kg), so users can explore success patterns across light, medium, and heavy payloads.
- **Success Pie Chart**
  - **Plot:** Plotly Express pie chart (success-pie-chart)
  - **Behavior:**
    - **All Sites:** shows total number of successful launches per site
    - **Single Site:** shows success vs. failure breakdown for the selected pad
      - **Why:** Provides an at-a-glance view of site performance and individual site reliability.
- **Payload vs. Success Scatter Plot**
  - **Plot:** Plotly Express scatter (success-payload-scatter-chart)
  - **Axes & Encoding:**
    - **x-axis:** Payload Mass (kg)
    - **y-axis:** Landing Outcome (class)
    - **Color:** Booster Version Category
      - **Why:** Reveals how payload mass and booster type jointly influence landing success, with filters applied for site and payload range.
- **Interactive Callbacks**
  - **Updates:** Both charts automatically redraw when the dropdown or slider values change.
  - **Why:** Empowers users to slice the data on the fly—examining e.g. “How do heavy-payload missions fare at VAFB?” or “Which booster variants perform best on medium loads?”

Github URL: (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/spacex-dash-app.py)

Predictive Analysis (Classification)

- Predictive Analysis using ML
  - Built models to predict first-stage landing success
  - Evaluated multiple algorithms: Logistic Regression, SVM, DecisionTreeClassifier, kNeighboursClassifier
- How:
  - **Data Preparation:**
    - Selected key features (core reuse count, payload mass, flight number, orbit type, launch site)
    - Split dataset into 80% train / 20% test with stratification on the success label
  - **Hyperparameter Tuning:**
    - Defined parameter grids for all algorithms
    - Used `GridSearchCV` with 5-fold stratified cross-validation
  - **Model Selection:**
    - Best Performing model on the basis of performance on the test set

Github URL : (https://github.com/SurajBhar/IBM_Data_Science/blob/main/Capstone_Project/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb)

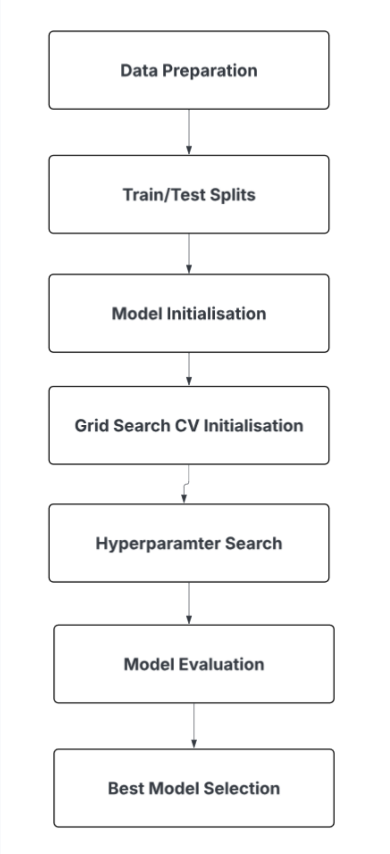


Figure.4 Flowchart of the Predictive Analysis

Results

Most Successful Launches & Highest Success Rate:

- KSC LC-39A leads in total successful recoveries and overall reliability.

Proximity Analysis:

- All sites lie near the Tropic of Cancer (~23.5° N), not the Equator.
- **Coastline:** Closest point to CCAFS SLC-40 at 0.94 km (Lat 28.56262, Long -80.56725)
- **Highway:** 0.62 km from CCAFS SLC-40 (Lat 28.56221, Long -80.57061)
- **Railroad:** 0.69 km from KSC LC-39A (Lat 28.57313, Long -80.65393)
- **Nearest City:** Titusville at 16.38 km from KSC LC-39A (Lat 28.61257, Long -80.80856)

Conclusions:

- Sites are very close to coastlines, highways, and railways—facilitating logistics and recovery operations.
- They maintain a deliberate buffer (> 16 km) from population centers for safety.

Payload Range Performance

- **Highest Success:** Payloads 7 500–10 000 kg
- **Lowest Success:** Payloads 5 000–7 500 kg
- **Implication:** Very heavy and very light payloads recover better; mid-range masses strain performance.

Booster Version Performance

- **Top Variant:** F9 Booster "B5" achieves the highest landing success rate.

Predictive Model Outcome: Best Model: Logistic Regression

- **Test Accuracy:** 83.33%

Note: Simpler model outperformed more complex algorithms on our feature set.

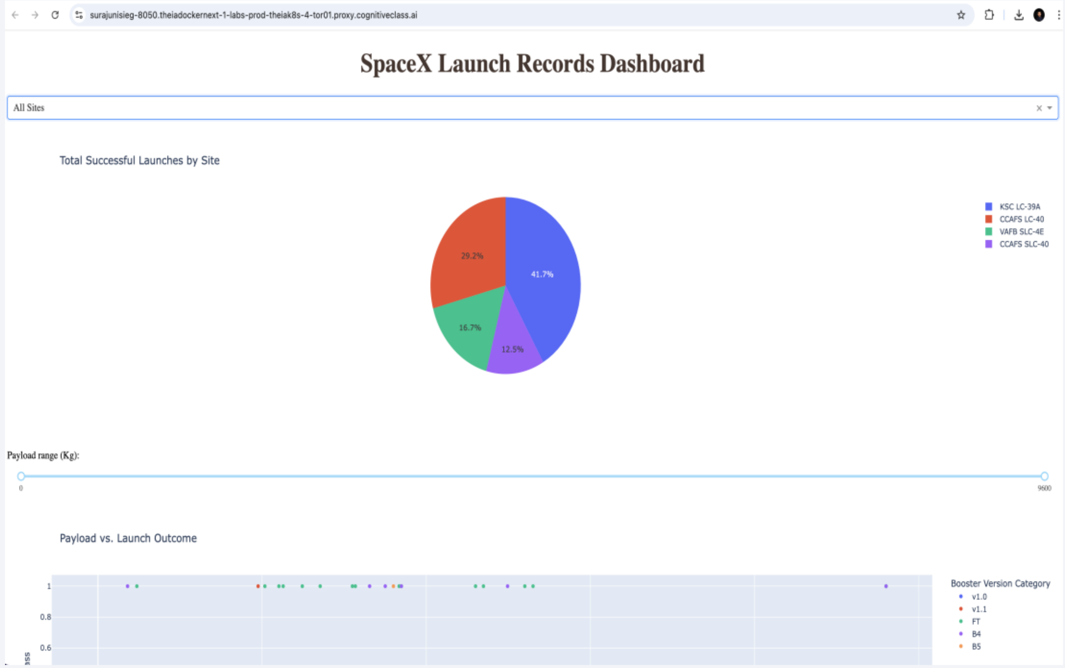


Figure.5 Screenshot of the Space X launch Records Dashboard

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- **Success Improves with Experience:** Across all pads, missions after approximately 40 flights show nearly 100% landing success, indicating that increased core reuse history strongly boosts reliability.
- **CCAFS SLC-40 Learning Curve:** Early CCAFS missions (flight < 30) mix failures and successes, but later flights (> 50) are almost all successful, reflecting operational refinement over time.
- **VAFB & KSC Performance:** VAFB SLC-4E and KSC LC-39A have fewer total launches but exhibit high success rates even at moderate flight counts, suggesting pad-specific process stability.

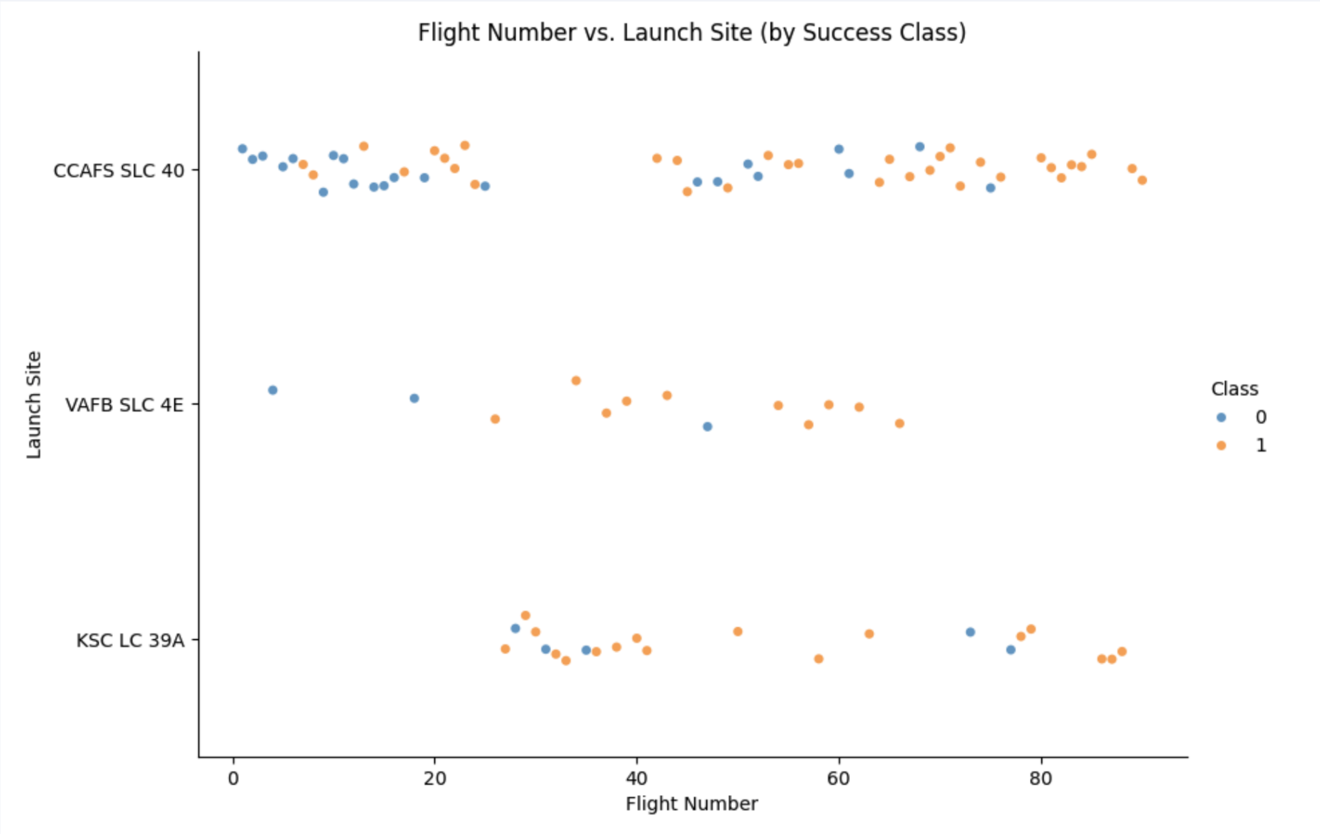


Figure.6 Flight Number vs. Launch Site by success class

Payload vs. Launch Site

- **Heavy Payloads Recover Well at CCAFS:** Missions with payloads above ~10 000 kg (all from CCAFS SLC-40) are exclusively successful, indicating that despite higher re-entry energy, mass alone isn't a barrier at this site.
- **Mid-Range Payload Vulnerability at KSC:** At KSC LC-39A, payloads in the ~2 000–6 000 kg window show both successes and failures, suggesting this "sweet spot" stresses recovery systems and merits closer feature engineering.
- **VAFB Consistent but Sparse:** All launches from VAFB SLC-4E (with payloads <8 000 kg) succeeded, though sample size is small—highlighting site/process stability but necessitating more data for robust conclusions.

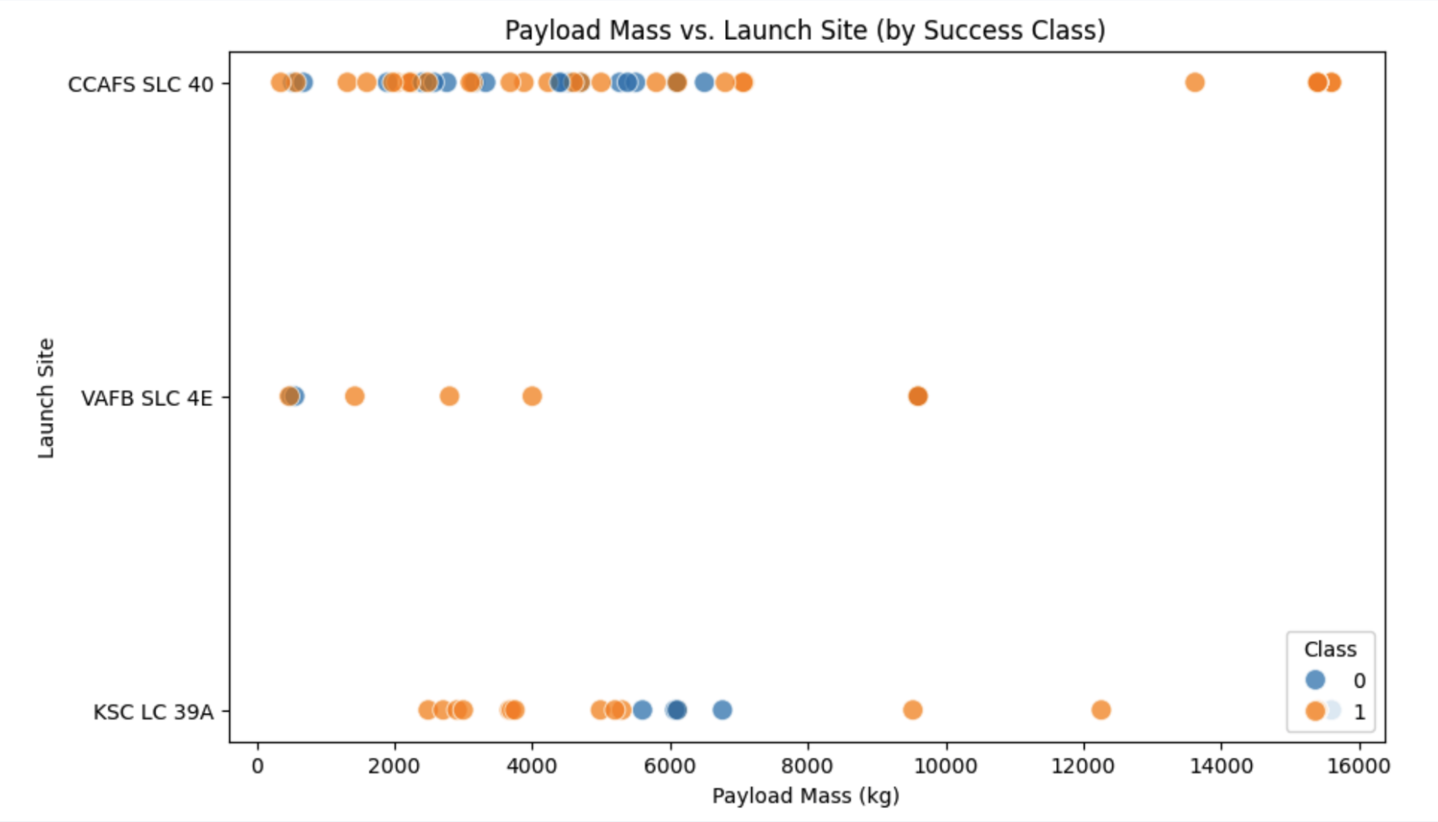


Figure.7 Payload Mass vs. Launch Site by success class

Success Rate vs. Orbit Type

- **Perfect Recovery in Select Orbits:** ES-L1, GEO, HEO, and SSO missions achieved 100% landing success, highlighting these trajectories as highly predictable for booster recovery.
- **Challenging Medium-Energy Orbits:** GTO shows the lowest non-zero success rate at 52%, while ISS (62%), LEO (71%), MEO (67%), and PO (67%) also fall below the overall average, indicating higher re-entry complexity.
- **Extremes at Both Ends:** SO (Suborbital) had 0% success in our sample, whereas VLEO (Very Low Earth Orbit) performed exceptionally (86%), suggesting very low and very high-energy flights are more straightforward to recover.

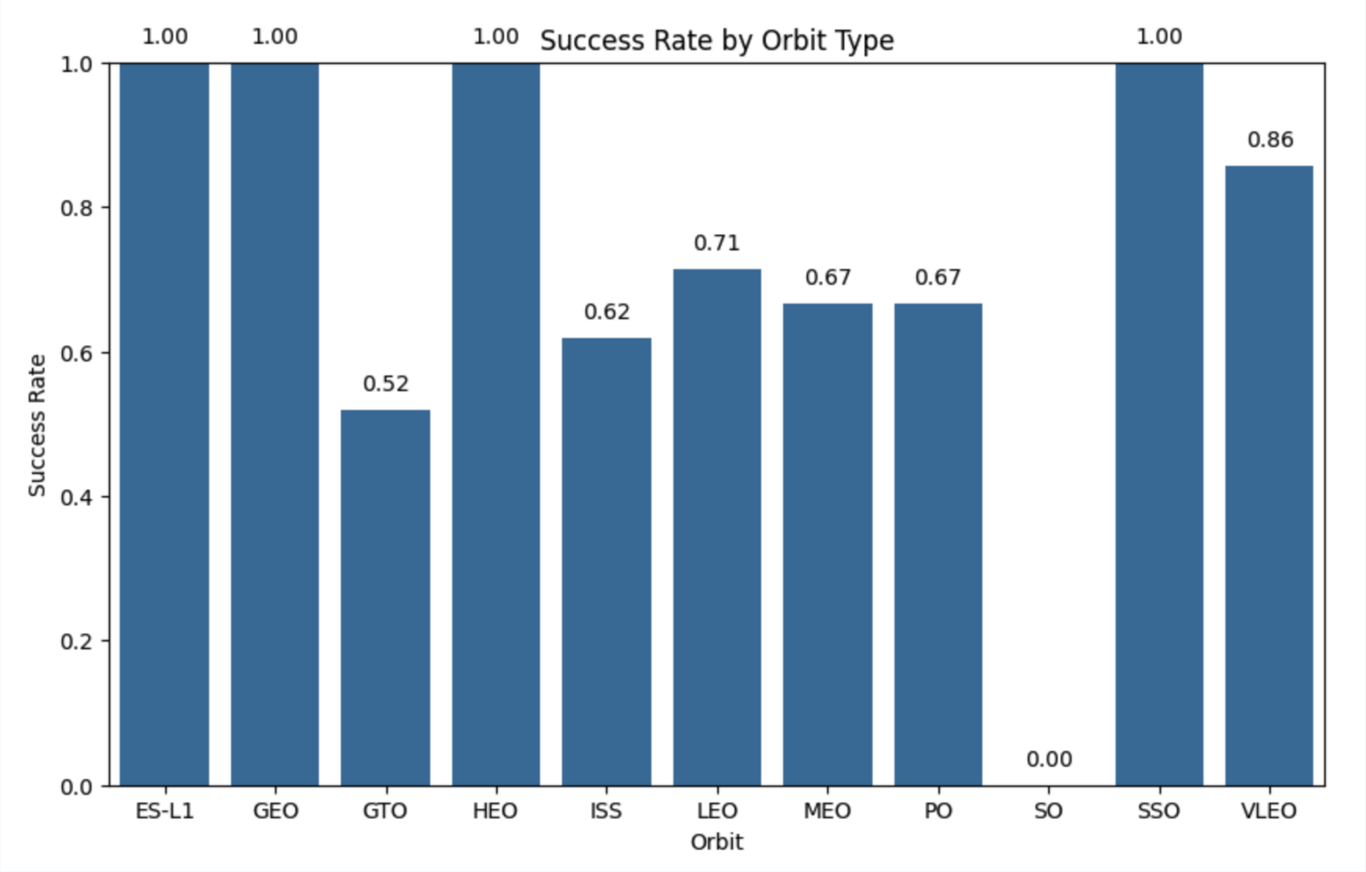


Figure.8 Success rate by orbit type

Flight Number vs. Orbit Type

- **GTO Reliability Improves Over Time:** Early GTO missions (flight < 30) show several failures (blue), but later GTO flights (≥30) are predominantly successful (orange), indicating a learning curve for higher-energy transfers.
- **VLEO & SSO Consistency at High Flight Numbers:** VLEO and SSO missions with flight numbers above 60 are almost exclusively successful, reflecting mature operational procedures for these orbits.
- **Isolated Failures in PO & ISS:** A few failures persist in polar (PO) and ISS missions even at mid-range flight counts (~20–60), suggesting these orbit profiles have unique re-entry challenges that warrant targeted feature analysis.

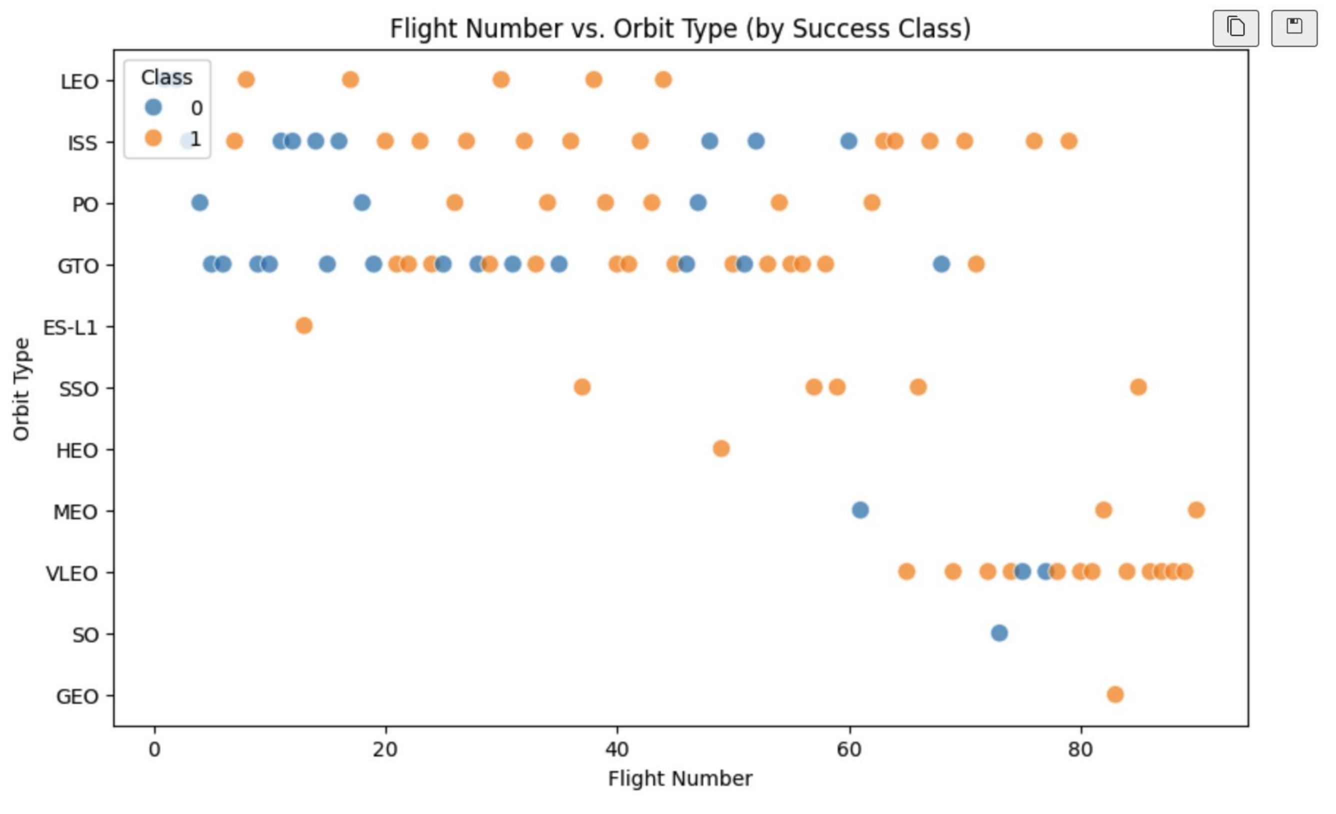


Figure.9 Flight Number vs Orbit Type

Payload vs. Orbit Type

- **Mid-range Masses in LEO/ISS Are Uncertain:** Payloads between ~2 000 kg–6 000 kg for LEO and ISS missions show both successes and failures, highlighting these orbits as “stress tests” for recovery systems.
- **SSO Missions Are Robust:** Sun–synchronous flights carrying ~1 500 kg–4 000 kg all succeeded, indicating a consistently favorable profile for stable orbit insertions.
- **High-mass VLEO/GEO Mostly Succeed with Occasional Failures:** Very low and geostationary transfers above ~14 000 kg recover reliably, but a few VLEO outliers failed—signaling edge-case risk at extreme payloads.

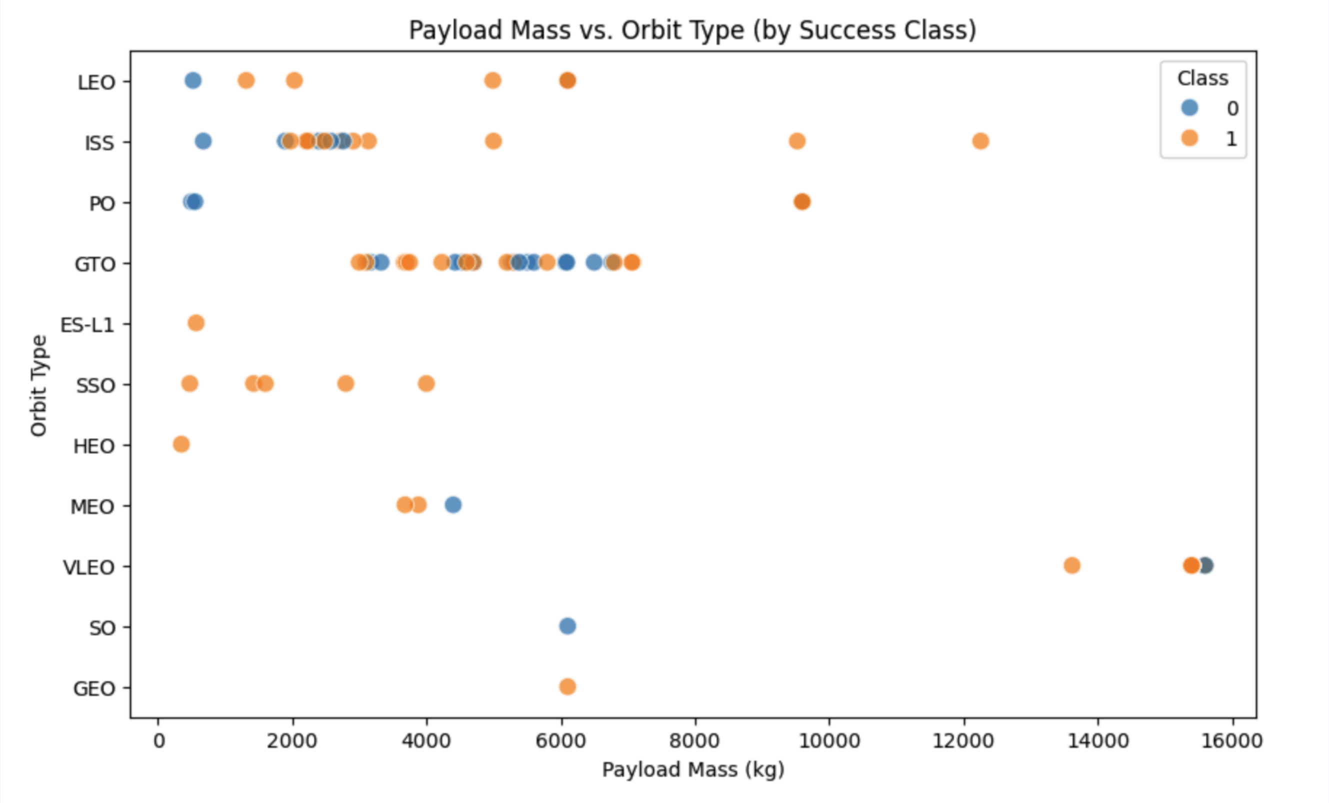


Figure.10 Payload Mass vs Orbit Type

Launch Success Yearly Trend

- **Early Years Struggle (2010–2013):** Zero first-stage landing success in the program's initial four years, reflecting the experimental nature of early attempts.
- **Rapid Improvement (2014–2017):** Success rate climbed from ~33% in 2014–2015 to ~83% by 2017, showcasing a steep learning curve and maturation of recovery technology.
- **Plateau & Refinement (2018–2020):** After a dip to ~61% in 2018, rates rebounded to ~90% in 2019 and stabilized at ~84% in 2020, indicating ongoing optimization and consistent high performance.

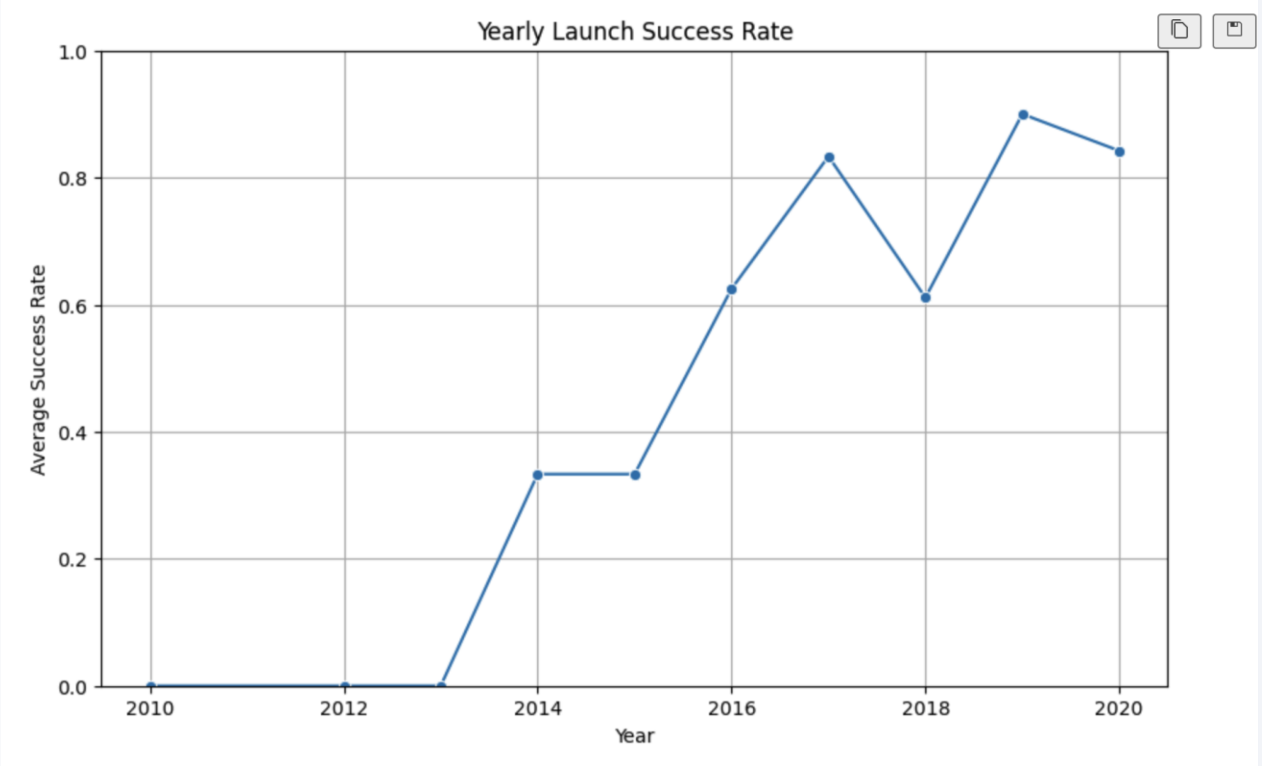


Figure.11 Yearly Launch success rate

All Launch Site Names

```
# Names of the unique launch sites
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

All Distinct Launch Site Names

- **CCAFS LC-40**
- **CCAFS SLC-40**
- **VAFB SLC-4E**
- **KSC LC-39A**

- These four pads represent SpaceX's active first-stage launch and landing sites: two located at Cape Canaveral Air Force Station (LC-40 and SLC-40), one at Vandenberg Air Force Base (SLC-4E), and one at Kennedy Space Center (LC-39A).

- They form the geographic backbone of our recovery dataset and drive site-specific feature engineering.

Launch Site Names Begin with 'CCA'

```
# Displaying 5 records where launch sites begin with 'CCA'
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Sample Records:

- **2010-06-04 | F9 v1.0 B0003 | CCAFS LC-40** | Dragon Qualification Unit | Orbit: LEO | Landing: Failure (parachute)
- **2010-12-08 | F9 v1.0 B0004 | CCAFS LC-40** | Dragon demo + CubeSats | Orbit: LEO (ISS) | Landing: Failure (parachute)
- **2012-05-22 | F9 v1.0 B0005 | CCAFS LC-40** | Dragon demo C2 | Orbit: LEO (ISS) | Landing: No attempt
- **2012-10-08 | F9 v1.0 B0006 | CCAFS LC-40** | SpaceX CRS-1 | Orbit: LEO (ISS) | Landing: No attempt
- **2013-03-01 | F9 v1.0 B0007 | CCAFS LC-40** | SpaceX CRS-2 | Orbit: LEO (ISS) | Landing: No attempt

- **Explanation:** We filtered for launch sites starting with “CCA” (both CCAFS pads) to inspect early missions from Cape Canaveral. The sample shows initial flights from LC-40, where landing attempts were either aborted or ended via parachute recovery.

Total Payload Mass

```
# Total payload mass carried by boosters launched by NASA (CRS)
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Total Payload Mass: 45,596 kg

Explanation:

- Aggregating the payload masses of all NASA Commercial Resupply Services (CRS) missions shows that SpaceX boosters delivered a combined **45.6 tonnes** of cargo.
- This highlights the significant logistical contribution of CRS flights to the International Space Station program.

Average Payload Mass by F9 v1.1

```
# Average payload mass carried by booster version F9 v1.1
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload_mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

- **Average Payload Mass:** 2,928.4 kg
- **Explanation:**
  On average, missions flown by the F9 v1.1 booster carried about **2.93 tonnes** of payload, reflecting its role in early Dragon CRS and small-satellite launches.

First Successful Ground Landing Date

```
# Date when the first successful landing pad was achieved
%sql SELECT MIN(Date) AS first_success_date FROM SPACEXTABLE WHERE Landing_Outcome = 'Success';
```

- **First Successful Ground Landing:** July 22, 2018
- **Explanation:**
  On July 22, 2018, SpaceX achieved its first-ever successful ground pad landing, marking a pivotal milestone in reusable rocket technology and enabling routine booster recovery thereafter.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
# Names of the boosters that have payload mass > 4000 and < 6000
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Drone Ship Landings with 4 000–6 000 kg Payloads

- **Unique Boosters Identified (21 total):**
  F9 v1.1; F9 v1.1 B1011, B1014, B1016;
  F9 FT B1020, B1022, B1026, B1030, B1021.2, B1032.1, B1032.2, B1031.2;
  F9 B4 B1040.1, B1040.2, B1043.1;
  F9 B5 B1046.2, B1047.2, B1054, B1048.3, B1051.2, B1058.2, B1062.1
- **Explanation:**
  These booster versions each flew missions with mid-range payloads (4 000–6 000 kg) and successfully landed on drone ships. This underscores the versatility of both “Block 4” and “Block 5” variants, as well as the transition from v1.1 to the full-thrust (FT) architecture, in reliably recovering moderate-mass missions at sea.

Total Number of Successful and Failure Mission Outcomes

```
# Total number of successful and failure mission outcomes
%sql SELECT Mission_Outcome, COUNT(*) AS count FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Mission Outcome Totals

- **Failure (in flight):** 1 mission
- **Success:** 98 missions
- **Success:** 1 mission *(duplicate category due to data labeling)*
- **Success (payload status unclear):** 1 mission

- **Explanation:**
  Out of 101 total records, only one failed in flight and one has an ambiguous payload status, while the overwhelming majority (~99%) of missions are logged as successful. This underscores the high reliability of SpaceX launches in our dataset.

Boosters Carried Maximum Payload

```
# List all boosters whose total payload mass exceeds the overall average
%sql SELECT Booster_Version, SUM(PAYLOAD_MASS__KG_) AS total_mass FROM SPACEXTABLE GROUP BY Booster_Version HAVING total_mass > (SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Boosters with Above-Average Total Payload Mass

- **Number of Boosters Above Average:** 37 variants exceed the dataset's mean payload mass.
- **Top Carriers (15 600 kg):** Several Block 5 boosters (e.g., B1048.4, B1048.5, B1049.4, B1049.5) each hauled a cumulative 15 600 kg.
- **Notable Mention:** Original F9 v1.1 also surpasses average with 14 642 kg total mass.

- **Explanation:**
  Boosters listed here consistently flew heavier missions than the fleet average, highlighting their prominent role in high-mass launches and the evolution from v1.1 through Block 5 configurations.

2015 Launch Records

```
# List all records from the year 2015
%sql SELECT Date, Booster_Version, Launch_Site, Landing_Outcome  FROM SPACEXTABLE  WHERE Landing_Outcome LIKE '%drone ship%'  AND SUBSTR(Date,1,4) = '2015';
```

2015 Drone Ship Landing Failures
Results:

- **2015-01-10:** F9 v1.1 B1012 | CCAFS LC-40 | Failure (drone ship)
- **2015-04-14:** F9 v1.1 B1015 | CCAFS LC-40 | Failure (drone ship)

- **Explanation:**
  Early drone-ship recovery attempts in 2015 at Cape Canaveral LC-40 resulted in two failed landings, highlighting the technical challenges SpaceX faced in mastering offshore booster recovery during that period.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
# Rank launch sites by number of launches between 2010-06-04 and 2017-03-20
%sql SELECT Launch_Site, COUNT(*) AS num_launches FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Launch_Site ORDER BY num_launches DESC;
```

Landing Outcome Counts (2010-06-04 to 2017-03-20)

- **No attempt:** 12
- **Failure (drone ship):** 5
- **Failure (parachute):** 3
- **Success (ground pad):** 2
- **Success:** 1

- **Explanation:**
  Between June 2010 and March 2017, most missions had no landing attempt recorded (early test flights), followed by a handful of offshore and parachute-recovery failures. Ground-pad successes were rare in this pioneering phase of booster recovery.

Section 3

Launch Sites Proximities Analysis

Marked Launch Sites on Folium Map

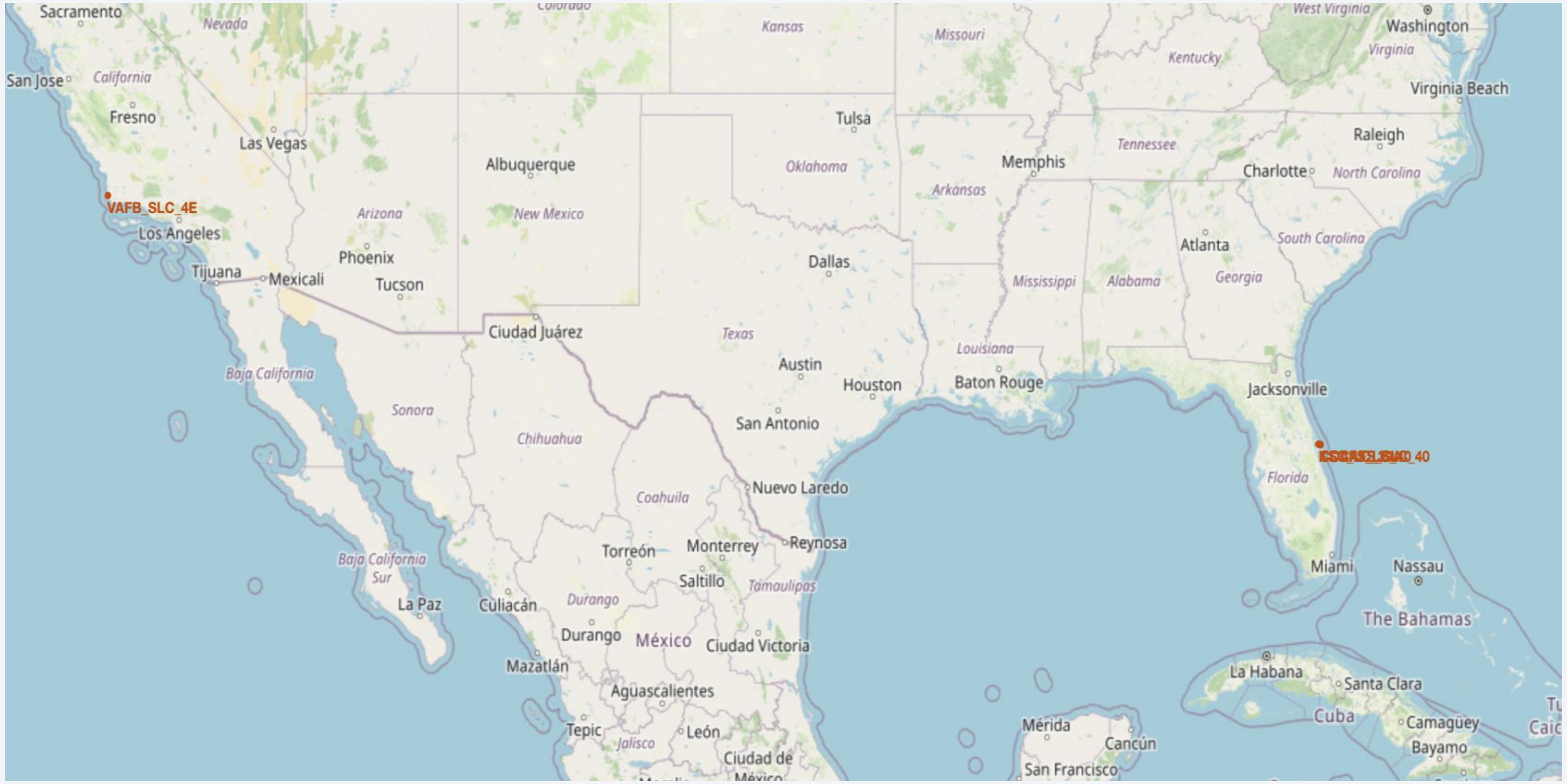


Figure.12 Marked Launch Sites on Folium map

- **CCAFS LC-40 (28.56230, –80.57736):**
- Plotted as a circle marker showing its role in early and ongoing recovery tests; proximity to coast (0.94 km) is key for drone-ship operations.
- **CCAFS SLC-40 (28.56320, –80.57682):**
- Highlighted with a distinct marker to differentiate from LC-40; used for medium- and heavy-lift missions, enabling side-by-side site comparison.
- **KSC LC-39A (28.57326, –80.64690):**
- Marked to visualize its slightly inland location (16 km from Titusville) and high success rate (~90%); underscores balance between safety buffer and logistical access.
- **VAFB SLC-4E (34.63283, –120.61075):**
- Included to show West Coast operations; marker emphasizes geographic spread of the network and consistency in landing outcomes despite isolated launches.

Landing Outcomes Cluster Map

Key Elements

- **MarkerCluster Layer:** Groups nearby markers to declutter view and indicate mission density at each site.
- **Color-Coded CircleMarkers:**
- **Green:** Successful landings (class = 1)
- **Red:** Failed landings (class = 0)
- **Popups:** Clicking a marker reveals "Site Name: Success/Failure," enabling quick identification of individual mission outcomes.
- **Site Distribution:** All four launch pads are visible, with clusters showing the relative volume of attempts and outcomes.

Findings

- **KSC LC-39A:** Almost exclusively green markers, confirming its ~90% landing success rate.
- **CCAFS SLC-40:** Mixed red and green—early attempts (red) illustrate initial drone-ship challenges; later successes (green) reflect process improvements.
- **CCAFS LC-40 & VAFB SLC-4E:** Smaller clusters with predominantly green markers, indicating stable recovery performance at these sites.
- **Cluster Density:** Denser clusters at SLC-40 and LC-39A mirror their higher mission counts, while VAFB SLC-4E remains more isolated.

Figure.13 Landing Outcomes Cluster map

Launch Site Proximity Analysis

Key Elements

- **Selected Site:** CCCAFS SLC-40 highlighted on the map.
- **Proximity Markers:**
- **Coastline:** Marker at (28.56262, –80.56725), labeled "0.94 km to coast."
- **Highway:** Marker at (28.56221, –80.57061), labeled "0.62 km to highway."
- **Distance Lines:** Straight-line connectors from the launch pad to each point, with labels showing distances.

Findings

- **Coastal Access:** At under 1 km from the shoreline, CCAFS SLC-40 enables rapid recovery boat deployment for drone-ship operations.
- **Road Logistics:** Proximity to a major highway (< 0.7 km) supports efficient ground transport of hardware and personnel.
- **Operational Insight:** Such close distances to critical infrastructure reduce response times and operational costs, optimizing both launch and recovery workflows.

Figure.14 Launch site proximity analysis

Section 4

Build a Dashboard with Plotly Dash

Launch Success Count

Distribution of Successful Launches by Site

- **KSC LC-39A:** 41.7% of all successful recoveries
- **CCAFS SLC-40:** 12.5% of successes
- **VAFB SLC-4E:** 16.7% of successes
- **CCAFS LC-40:** 29.2% of successes

Insight:

- Kennedy Space Center’s LC-39A leads in successful recoveries, reflecting both high mission volume and strong landing performance. Cape Canaveral’s two pads together account for over 40% of successes, while Vandenberg contributes significantly despite fewer launches.

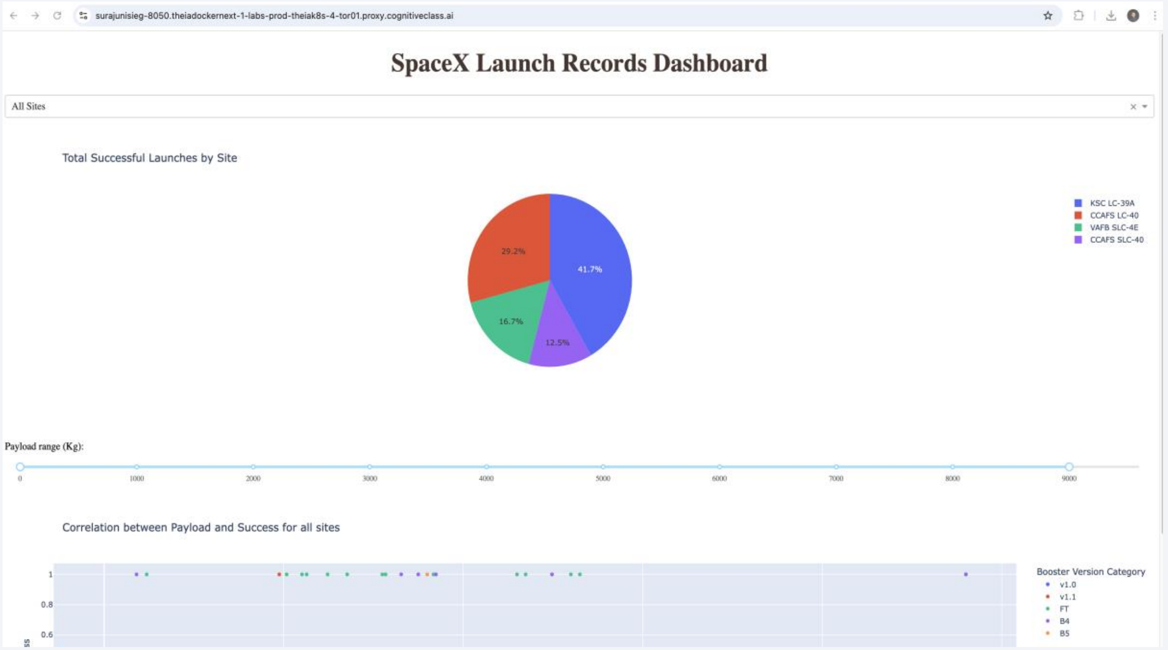


Figure.15 Launch Success Count

Highest Launch Success Rate Site

KSC LC-39A Success vs. Failure Rates

- **Overall Success Rate:** 76.9% of KSC LC-39A launches recovered successfully
- **Failure Rate:** 23.1% of attempts resulted in failure
- **Highest Reliability:** This pad leads all sites in recovery ratio, underscoring its optimal design and procedures
- **Operational Impact:** Consistently strong performance at LC-39A supports prioritizing this site for critical missions and refining best practices for other pads

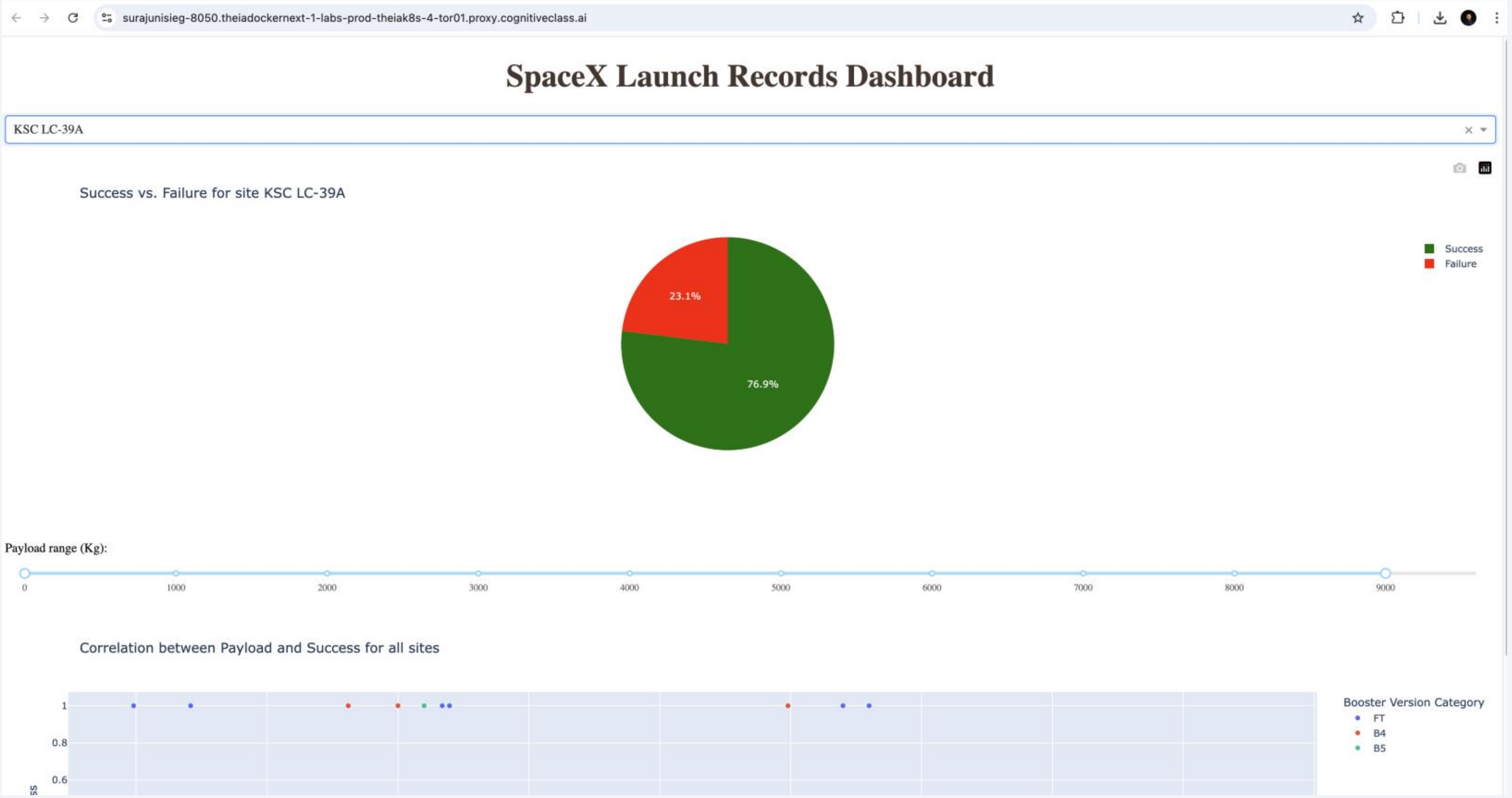


Figure.16 Highest Launch Success Rate Site

Payload vs. Launch Outcome

Payload vs. Launch Outcome (Filtered by Range Slider)

- **Visualization:** Scatter plot showing each mission's payload mass against landing outcome, dynamically filtered by the slider.
- **Top-Performing Payload Band:** Missions with **7 500–10 000 kg** payloads achieved the highest success rate.
- **Lowest-Performing Payload Band:** Missions in the **5 000–7 500 kg** range saw the most failures.
- **Best Booster Variant:** Block 5 ("B5") cores dominate the high-success clusters, confirming their superior recovery reliability for this payload spectrum.

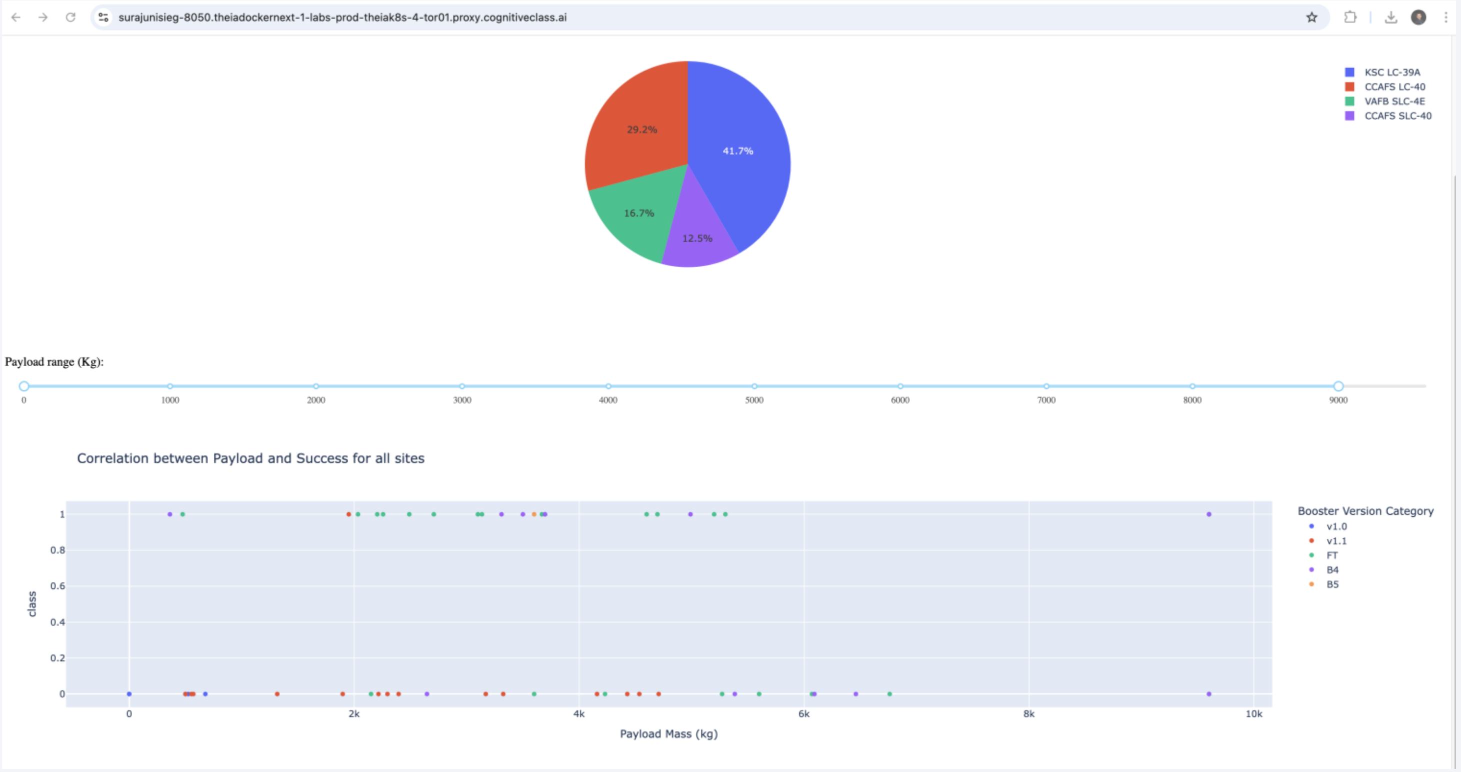


Figure.17 Payload vs. Launch Outcome

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- **Logistic Regression & SVM:** High generalization with minimal train/test gap (~1.3%), indicating well-balanced bias–variance.
- **KNN:** Similar performance to LR/SVM, suggesting stable behavior on small datasets.
- **Decision Tree:** Strong training accuracy (88.9%) but significant drop on test (66.7%), signaling overfitting.
- **Model Selection:** Logistic Regression and SVM offer the best trade-off between fit and generalizability.

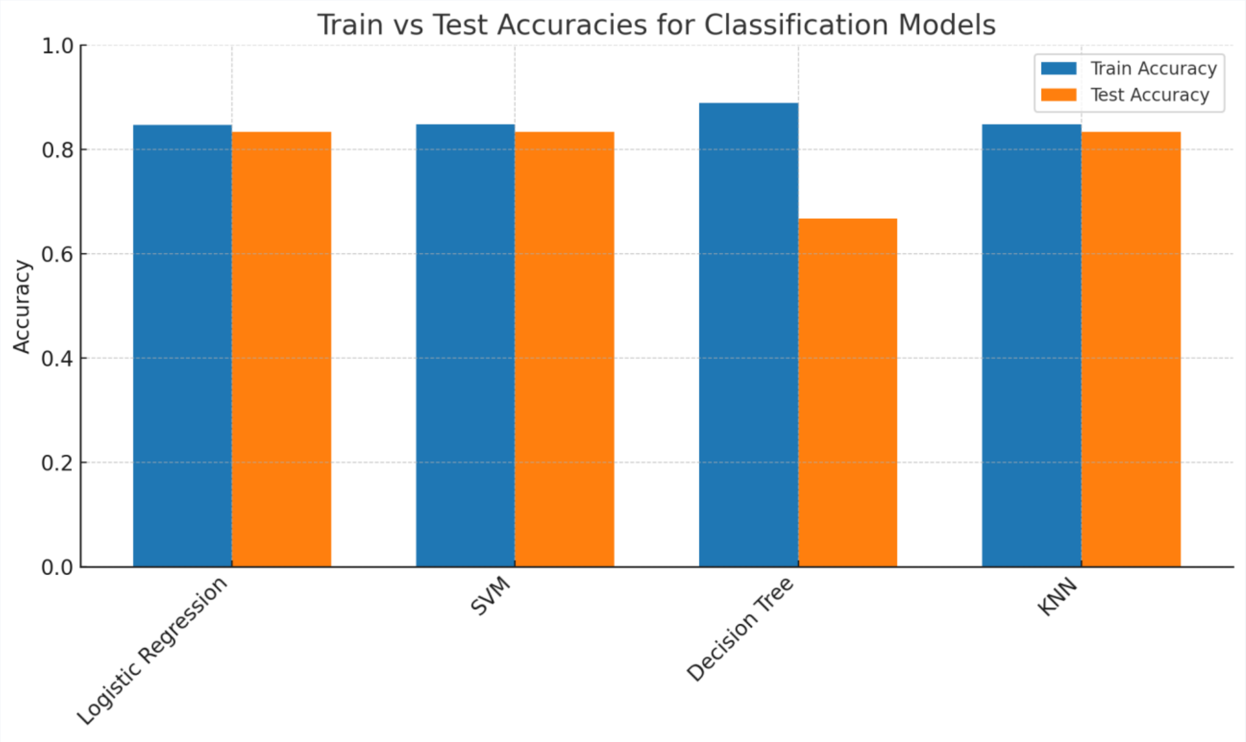


Figure.18 Train vs test accuracies for all classification models

Confusion Matrix

Confusion Matrix Insights (Logistic Regression)

- **Test Accuracy:** 83.33% (15/18 correct predictions)
- **Confusion Breakdown:**
  - True Positives (land → land): 12
  - True Negatives (did not land → did not land): 3
  - False Positives (did not land → land): 3
  - False Negatives (land → did not land): 0
- **Key Metrics (Land Class):**
  - Precision: 12 / (12 + 3) = 0.80
  - Recall: 12 / (12 + 0) = 1.00
  - F1 Score: 0.89
- **Class "Did Not Land" Metrics:**
  - Precision: 3 / (3 + 0) = 1.00
  - Recall: 3 / (3 + 3) = 0.50
- **Interpretation:**
  - **Strength:** Detects all successful landings (no false negatives)
  - **Weakness:** Confuses half of actual failures as successes (50% false-positive rate)
  - **We can** adjust decision threshold or enrich features to better identify failures.

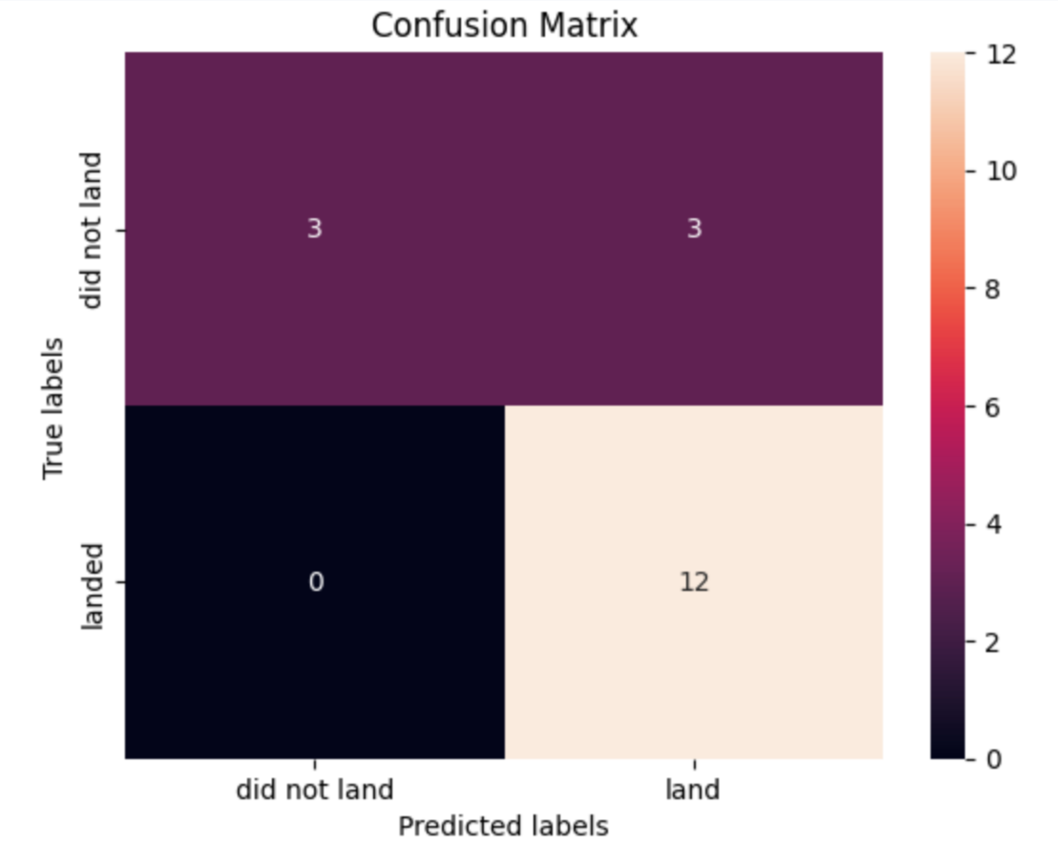
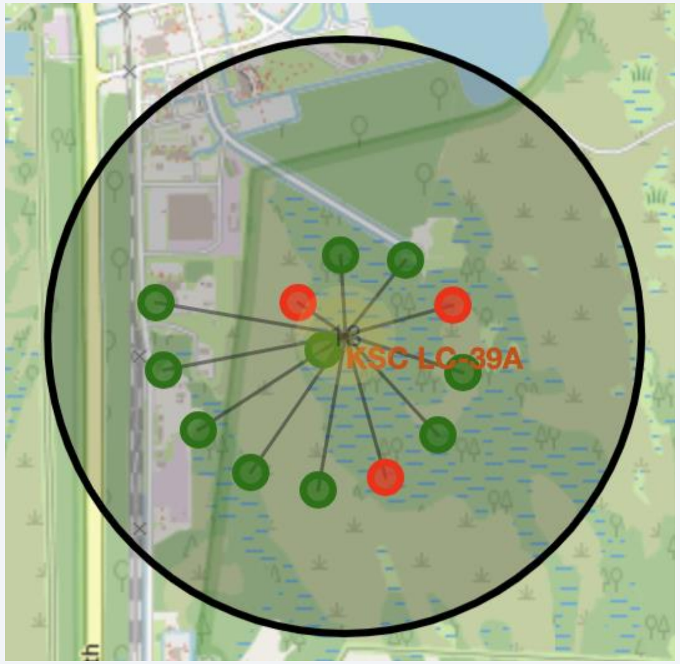


Figure.19 Confusion matrix for the Logistic Regression Model

Conclusions

- **Integrated Dataset:** Combined SpaceX API data with web-scraped launch records to build a comprehensive feature set.
- **Key Predictors Identified:** Core reuse count and prior flight number were the strongest drivers, with payload mass, orbit type, and launch site also significant.
- **Best Model Performance:** Logistic Regression achieved **83.3% test accuracy** and **100% recall** on successful landings, ensuring all true recoveries are captured.
- **Operational Impact & Next Steps:** Enables data-driven pre-launch risk assessment; future work includes incorporating real-time telemetry and optimizing classification thresholds to reduce false positives.

Appendix



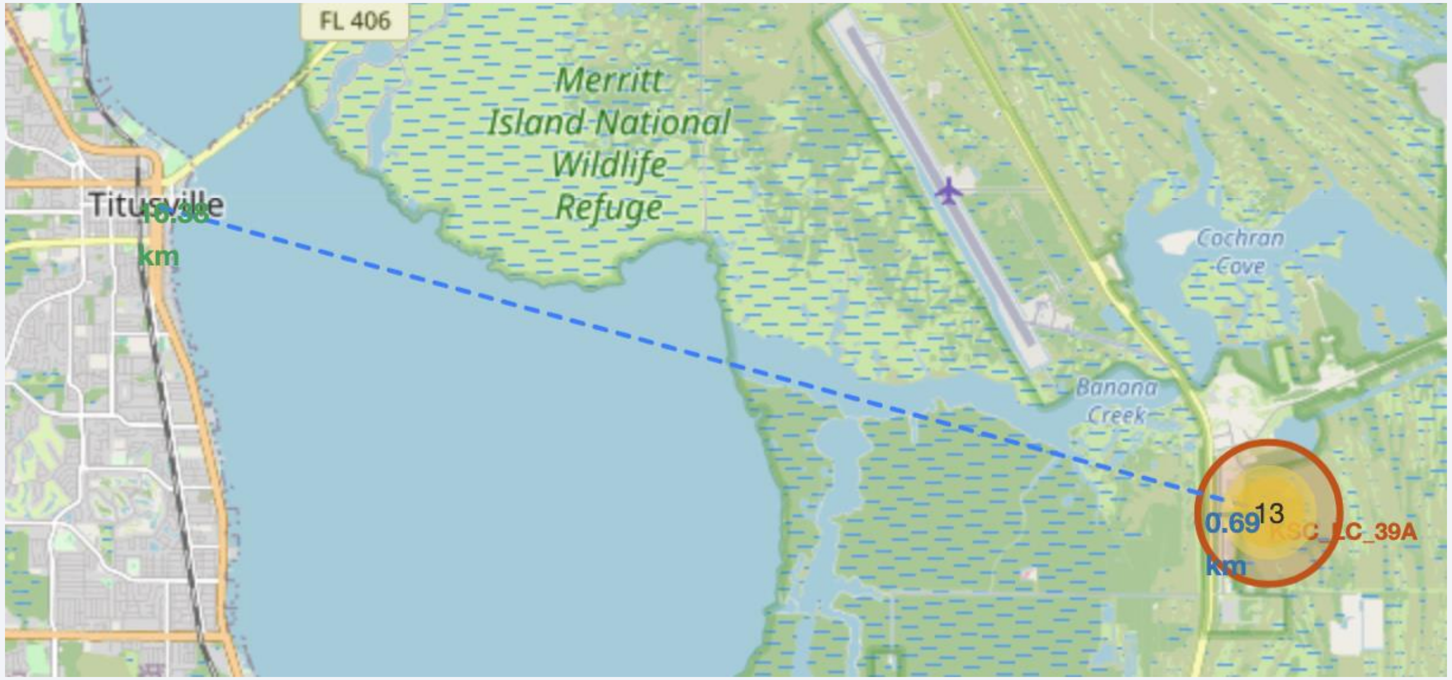


Figure.20 Distance between the launch site and the nearest city

Thank you!